

Section A

Q.1 (a) Define Bracket and signum function with graphical representation, also discuss the discontinuity.

(b) Evaluate (i) $\lim_{x \rightarrow 3^+} \frac{3-x}{|x-3|}$ 

on Right side: only numerator change with - outside the bracket, and LEFT side, only numerator change without - outside the bracket

Q.2 (a) Find constant a and b such that the function f defined by

$$\begin{aligned} \text{let } f(x) &= x^2 & \text{if } x < 1 \\ &= ax + b & \text{if } -1 \leq x \\ &= x^2 + 2 & \text{if } x \geq 1. \end{aligned} \quad \text{Is cont for all x.}$$

(b) Discuss the continuity of $\frac{\sqrt{1-x^2}}{1-x}$ at $x = 1$

Q.3 Find dy/dx , $y = x^x + \ln x^x$ **Q.4 Find $\frac{d^2u}{dx^2}, \frac{d^2u}{dy^2}$ $u = e^{x^y}$**

Q.5 Find the value of c (i) by Rolle's theorem if $f(x) = \cos x$ $[\pi/2, 3\pi/2]$. (ii) by M.V.T if $f(x) = x^3 - 3x^2 + 2x$ on $[0, 1/2]$

Q.6 Solve by left hospital rule (a) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$ $a > 1$

Q.4 (a) If $f(x) = \ln(1 + \sqrt{1-x})$ prove that $4x(1-x)f''(x) + 2(2-3x)f'(x) + 1 = 0$.

Q.2 Find dy/dx . $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$

Section B

Q.7 $\int \frac{x^2+x+2}{(3x-1)(x^2+1)} dx$ **Q.8 $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx$** **Q.9 $\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}}$** **Q.10 Evaluate $\int \frac{1-\sin x}{1-\cos x} e^x dx$**

(a) $\int e^{ax} \cos(bx+c) dx$ (b) $\int_0^{\pi/2} \sin^2 x / 2 \cos^2 x / 2 dx$ (c) $\int \frac{\ln x}{\sqrt{x}} dx$

(d) $\int x^2 \sec^2 x^2 dx$ (e) $\int_{\sqrt{e}}^e \frac{\ln x}{x^2} dx$

Evaluate $\int_{-1}^5 |x-2| dx$ $\int \frac{4e^x + 6e^{-x}}{9e^x + 4e^{-x}} dx$

Q.1 (a) Find the equation of the straight line passing through the point (3, 4, 5) and intersecting the z axis at right angles.

(b) Find the equation of the plane passing through the points (2, -3, 2) and (1, 2, -1) and is Perpendicular to the plane $3x - 2y + z = 7$.

(c) Find the equation of the plane which is perpendicular bisector of the segment joining the point (3, -2, 5) and (5, 4, -3).

Q.2 (a) Find the equation of the line passing through the point (2, -3, 2) and perpendicular to each of the line $5x = y + 3 = 2 - z$ and $x + 3 = 1 - 2y = 1 + z$.

(b) Find the equation of the plane containing the lines

$$\frac{x+2}{2} = \frac{3-y}{3} = \frac{z}{2} \text{ and } \frac{x+2}{3} = \frac{y-3}{2} = \frac{z}{1}.$$

(c) Show that the line $\frac{x+1}{2} = \frac{y-2}{2} = \frac{z-5}{-3}$ and the line $\frac{x-3}{-1} = \frac{y-2}{-3} = \frac{z}{2}$ are intersect and

find the coordinates of the common point.

30. The plane through the points $P_1(-2, 1, 4)$, $P_2(1, 0, 3)$ that is perpendicular to the plane $4x - y + 3z = 2$.

31. The plane through $(-1, 2, -5)$ that is perpendicular to the planes $2x - y + z = 1$ and $x + y - 2z = 3$.

32. The plane that contains the point $(2, 0, 3)$ and the line $x = -1 + t$, $y = t$, $z = -4 + 2t$.

33. The plane whose points are equidistant from $(2, -1, 1)$ and $(3, 1, 5)$.

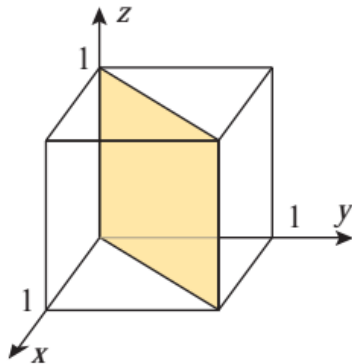
25–26 Find the intersections of the lines with the xy -plane, the xz -plane, and the yz -plane. ■

25. $x = -2$, $y = 4 + 2t$, $z = -3 + t$

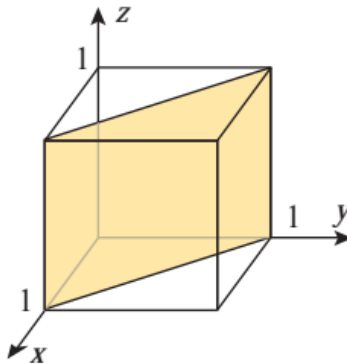
26. $x = -1 + 2t$, $y = 3 + t$, $z = 4 - t$

7–10 Find an equation of the plane indicated in the figure. ■

7.



8.



22. The line through the origin that is parallel to the line given by $x = t$, $y = -1 + t$, $z = 2$.

23. Where does the line $x = 1 + 3t$, $y = 2 - t$ intersect

(a) the x -axis

(b) the y -axis

(c) the parabola $y = x^2$?

24. Where does the line $\langle x, y \rangle = \langle 4t, 3t \rangle$ intersect the circle $x^2 + y^2 = 25$?